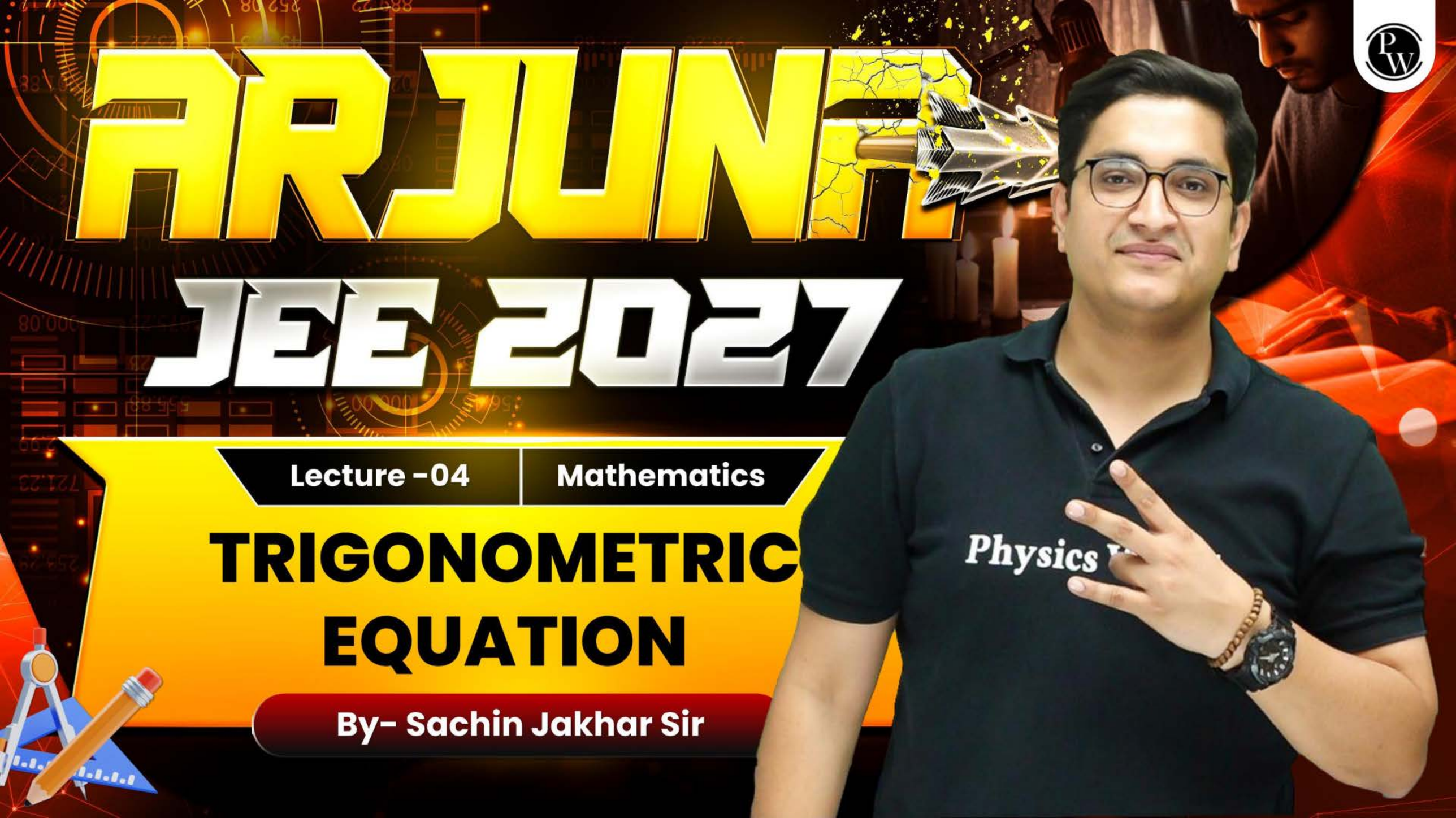
ARJUNA JEE 2027

Lecture -04 | Mathematics

TRIGONOMETRIC EQUATION

By- Sachin Jakhar Sir



Topics *to be covered*

A **Types of Trigonometric Equations**

B **Question Practice**

C

D



H.W. Discussion

Question

$(2^x + 2^{-x}), (f(x)), (3^{1-x} + 3^{1+x})$ are in A.P. Then find minimum value of $f(x)$.

$$f(x) = \left(\frac{2^x + 2^{-x} + 3^{1-x} + 3^{1+x}}{2}\right)$$

$$(AM \geq GM) \longrightarrow \frac{2^x + 2^{-x} + 3^{1-x} + 3^{1+x}}{4} \geq \left(2^x \cdot 2^{-x} \cdot \underbrace{3^{1+x} \cdot 3^{1-x}}_{\downarrow 9}\right)^{\frac{1}{4}}$$

$$\frac{2f(x)}{4} \geq \sqrt{3}$$

$$f(x) \geq 2\sqrt{3} \quad \text{X (wrong).}$$

equality hold

$$\underbrace{2^x}_{\downarrow x=0} = \underbrace{2^{-x}}_{x=\text{different}} = \underbrace{3^{1+x}}_{\downarrow x=0} = 3^{1-x}$$

$$f(x) = \left(\frac{2^x + 2^{-x} + 3^{1-x} + 3^{1+x}}{2}\right)$$

$(AM \geq GM) \longrightarrow \quad \frac{2^x + 2^{-x}}{2} \geq (1)^{\frac{1}{2}} \Rightarrow 2^x + 2^{-x} \geq 2 \quad (\text{when } x = 0)$

$(AM \geq GM) \longrightarrow \quad \frac{3^{1+x} + 3^{1-x}}{2} \geq (9)^{\frac{1}{2}} \Rightarrow 3^{1+x} + 3^{1-x} \geq 6 \quad (\text{when } x = 0)$

$$f(x) = \frac{1}{2}\left(\underbrace{2^x + 2^{-x}}_{\downarrow \min = 2} + \underbrace{3^{1-x} + 3^{1+x}}_{\downarrow \min. = 6}\right) = \frac{1}{2}(2+6) = \boxed{4}$$

both at $x = 0$

for

Question [JEE Mains-2020]

70%

The number of distinct solutions of the equation, $\log_{1/2}|\sin x| = 2 - \log_{1/2}|\cos x|$ in the interval $[0, 2\pi]$, is

$x \in [0, 2\pi]$

$2x \in [0, 4\pi]$

$$\log_{\frac{1}{2}}|\sin x| + \log_{\frac{1}{2}}|\cos x| = 2$$

$$\left(|\sin x \cdot \cos x|\right) = \left(\frac{1}{2}\right)^2 = \frac{1}{4}$$

$$\left|\frac{\sin 2x}{2}\right| = \frac{1}{4_2}$$

$$|\sin 2x| = \frac{1}{2}$$

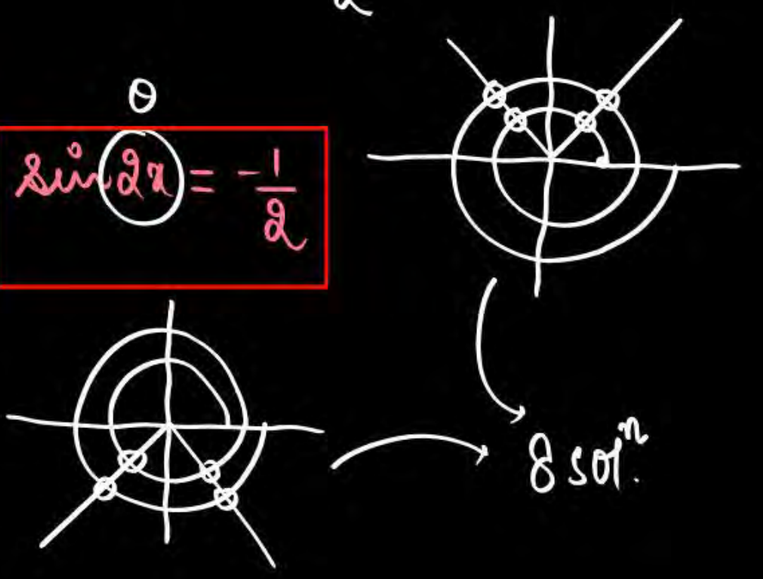
θ

$\sin(2x) = \frac{1}{2}$

$\sin\theta = \frac{1}{2}$, $\theta \in [0, 4\pi]$

θ

$\sin(2x) = -\frac{1}{2}$



RAPID FIRE

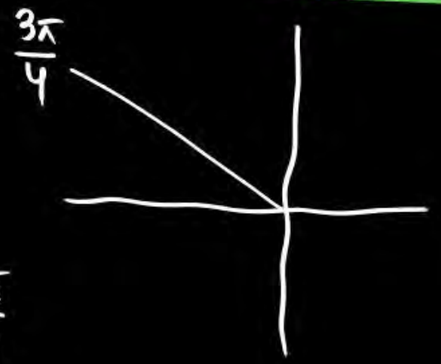
Solutions in case if two equations are given:

1. If $\sin(\theta) = \frac{1}{\sqrt{2}}$ and $\cos(\theta) = -\frac{1}{\sqrt{2}}$, then find the general values of θ.
2. If $\sin(\theta) = \frac{1}{\sqrt{2}}$ and $\tan(\theta) = -1$, then find the general values of θ.
3. If $\cos\theta = \frac{1}{\sqrt{2}}$ and $\tan\theta = -1$, then find the general value of θ.

$\theta \in$ II quad.

$\theta = \pi - \frac{\pi}{4} = \frac{3\pi}{4}$

$\therefore \theta = \left\{2n\pi + \frac{3\pi}{4}\right\}, n \in I.$



$\theta \in$ IV quad.

$\theta = 2\pi - \frac{\pi}{4}$ OR $-\frac{\pi}{4}$

$= \frac{7\pi}{4}$

$\left\{2n\pi + \frac{7\pi}{4}\right\}$ OR $\left\{2n\pi - \frac{\pi}{4}\right\}$

Same.

Question

If $\cos\theta = -\frac{1}{\sqrt{2}}$ and $\tan\theta = 1$, then the general value of θ is

A $2n\pi + \frac{\pi}{4}$

B $(2n+1)\pi + \frac{\pi}{4}$

C $n\pi + \frac{\pi}{4}$

D $n\pi \pm \frac{\pi}{4}$

$\theta \in$ III quad.

$\theta = \pi + \frac{\pi}{4}$, $-\pi + \frac{\pi}{4}$

add $2n\pi$

[where n ∈ I],

Challenger Question

Principal solution of equation : $\sin^2 x + 2\tan^2 x + \frac{4}{\sqrt{3}}\tan x - \sin x + \frac{11}{12} = 0$



Question [JEE (Adv.)-2018]

Let a, b, c be three non zero numbers such that the equation $\sqrt{3}a\cos x + 2b\sin x = c$, $x \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ has two distinct real roots α and β with $\alpha + \beta = \frac{\pi}{3}$. Then the value of $\frac{b}{a}$ is

Type - 2

$\sqrt{3}\,a\cos x + 2b\sin x = c$ — α, β.

HW

Question

If $\tan\theta = \cot 2\theta$, then the value of θ is

A $(2n+1)\frac{\pi}{6}; n \in I$

B $(2n+1)\frac{\pi}{6}; n \in I; n \neq 3m+1; m \in I$

C $(2n+1)\frac{\pi}{6}; n \in I; n \neq 2m+1; m \in I$

D none

$\tan\theta$

$\theta \neq \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2} \ldots\ldots$

$\cot(2\theta)$

$\theta \neq \frac{\pi}{2}, \frac{2\pi}{2}, \frac{3\pi}{2}, \frac{4\pi}{2} \ldots$

$\pi \quad 2\pi$

$\tan\theta = \tan\left(\frac{\pi}{2} - 2\theta\right)$

$\theta = n\pi + \frac{\pi}{2} - 2\theta$

$3\theta = \frac{2n\pi + \pi}{2}$

$\theta = \frac{(2n+1)\pi}{6}, \quad n \in I - \{1, 4, 7, 10, 13 \ldots\}$

Same: $3m+1$ OR $3m-2$

$1 + (m-1)3$

$n=1 \quad n=4 \quad n=7$

$\frac{\pi}{6}, \frac{3\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{9\pi}{6} \ldots\ldots \frac{15\pi}{6}$

$\frac{\pi}{2} \quad \frac{3\pi}{2} \quad \frac{5\pi}{2}$

Union

$\cup$

OR

{जहां दोनों में कोई एक भी हो जाए तो चलेगा}

$\theta = n\pi + (-1)^n \alpha$ $\begin{cases} n \to \text{odd} \\ n \to \text{even} \end{cases} \cup$

$\theta = 2n\pi \pm \alpha$ $\begin{cases} + \\ - \end{cases} \cup$

Intersection

$\cap$

AND

{जहां दोनों का एक साथ होना जरूरी हो}

$(\quad)^2 + (\quad)^2 = 0$

$\underline{=} 0 \qquad \underline{=} 0$



DHYAAN RKHNA!!

4. Aapka answer, dikhne mein alag ho skta hai.

$$\sin x = -\frac{1}{2}$$

IV

$\sin x = \sin\left(-\frac{\pi}{6}\right)$

$\Downarrow$

$$x = n\pi + (-1)^{n+1}\frac{\pi}{6}$$ ✓

$\sin x = \sin\left(\pi + \frac{\pi}{6}\right)$

$\Downarrow$

$$x = n\pi + (-1)^{n}\frac{7\pi}{6}$$ ✓

exactly same.

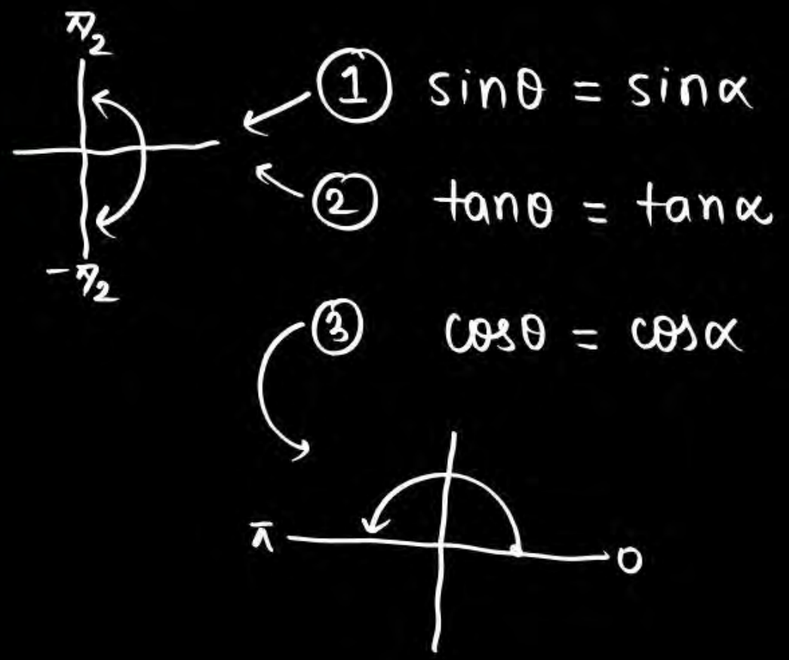
IMPORTANT NOTE

sin, cos, tan → +ve value then $\alpha \in$ I Quadrant.

sin → −ve value then take $\alpha \in$ IV Quadrant.

cos→ −ve value then take $\alpha \in$ II Quadrant.

tan → −ve value then take $\alpha \in$ IV Quadrant.



THOUGHT PROCESS

Try to make function & argument (angle) same.

Factorize into linear factors and write general solution.

Agr factorization na ho to "perfect square" bano.

Also look for an identity.

$a^3+b^3+c^3 = 3abc$

$a = b = c$

OR

$a+b+c = 0$



TYPES OF TRIGONOMETRIC EQNs.

Type-I : Solutions of Equations by Factorising:

Solution of trigonometric equation by factorization

Equation which are expressed in quadratic form

100% Ques^{n} आएगा

Question

Solve : $(2\sin x - \cos x)(1 + \cos x) = \sin^2 x$ **in** $[0, 2\pi]$.

$$(2\sin x - \cos x)(1 + \cos x) - (1^2 - \cos^2 x) = 0$$

$$(1 - \cos x)(1 + \cos x)$$

$$(1 + \cos x)(2\sin x - \cos x - 1 + \cos x) = 0$$

$\cos x = -1$

$x = \pi$

$\sin x = \frac{1}{2}$

$x = \frac{\pi}{6}, \frac{5\pi}{6}$

$ab = 0$

$a = 0$ OR $b = 0$

Ans. $\Rightarrow$ $x = \frac{\pi}{6}, \frac{5\pi}{6}, \pi$

Question

8 Ans.

Find the number of solution of the equation $16^{\sin^2 x} + 16^{\cos^2 x} = 10$, where $0 \le x \le 2\pi$.

$16^{\cos^2 x} = 16^{1-\sin^2 x}$

$= \dfrac{16^1}{16^{\sin^2 x}}$

$16^{\sin^2 x} + \dfrac{16}{16^{\sin^2 x}} = 10$

put # $16^{\sin^2 x} = t$

$t + \dfrac{16}{t} = 10$

$t^2 + 16 = 10t$

$t^2 - 10t + 16 = 0$

$-8t \quad -2t$

$(t-8)(t-2) = 0$

$t = 8$ OR $t = 2$

$16^{\sin^2 x} = 8$ $\qquad$ $16^{\sin^2 x} = 2$

$2^{4\sin^2 x} = 2^3$ $\qquad$ $2^{4\sin^2 x} = 2$

$t = 8$ OR $t = 2$

$16^{\sin^2 x} = 8$

$2^{4\sin^2 x} = 2^3$

$4\sin^2 x = 3$

$\sin^2 x = \frac{3}{4}$

$\sin x = \pm \frac{\sqrt{3}}{2}$ → 4 soln.

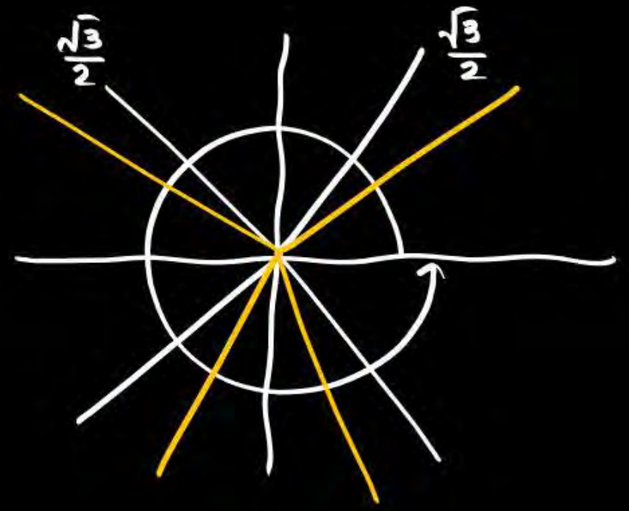
$16^{\sin^2 x} = 2$

$2^{4\sin^2 x} = 2^1$

$4\sin^2 x = 1$

$\sin^2 x = \frac{1}{4}$

$\sin x = \pm \frac{1}{2}$ → 4 soln.



Question [JEE Mains-2021]

The number of roots of the equation $(81)^{\sin^2 x} + (81)^{\cos^2 x} = 30$ in the interval $[0, \pi]$ is equal to:



80% QIBY

A 3

B 4

C 8

D 2

Question [JEE Mains-2023]

Let $S = \left\{x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) : 9^{1-\tan^2 x} + 9^{\tan^2 x} = 10\right\}$ and $\beta = \sum_{x \in S} \tan^2\left(\frac{x}{3}\right)$

then $\frac{1}{6}(\beta - 14)^2$ is equal to

Say: $9^{\tan^2 x} = t$

HW

A 16

B 32

C 8

D 64

Question [JEE Mains-2022]

Let $S = \left\{\theta \in [-\pi, \pi] - \left\{\pm\frac{\pi}{2}\right\} : \sin\theta\tan\theta + \tan\theta = \sin 2\theta\right\}$.

If $T = \sum_{\theta\in S}\cos 2\theta$, then $T + n(S)$ is equal to

A $7+\sqrt{3}$

B 9

C $8+\sqrt{3}$

D 10

$$\frac{\sin\theta}{\cos\theta}(\sin\theta + 1) - 2\sin\theta\cos\theta = 0$$

$$\sin\theta\left\{\frac{\sin\theta + 1}{\cos\theta} - 2\cos\theta\right\} = 0$$

$$\sin\theta\left(\frac{\sin\theta + 1 - 2\cos^2\theta}{\cos\theta}\right) = 0$$

$$\sin\theta\left(\frac{\sin\theta+1-2\cos^2\theta}{\cos\theta}\right)=0$$

$\sin\theta=0$

$$\sin\theta+1-2\cos^2\theta=0$$

$$\sin\theta+1-2(1-\sin^2\theta)=0$$

$$2\sin^2\theta+\sin\theta-1=0$$

$$(2\sin\theta-1)(\sin\theta+1)=0$$

$\sin\theta=\frac{1}{2}$ $\qquad$ $\sin\theta=-1$

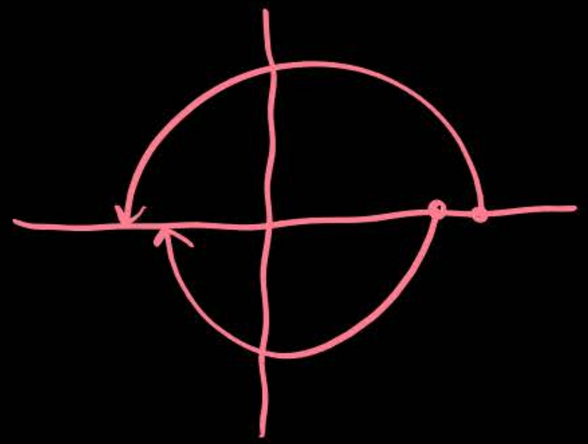
$$2t^2+t-1=0$$

$$2t^2+2t-t-1=0$$

$$(2t-1)(t+1)=0$$

$\theta \in [-\pi, \pi]$

$[-\pi, 0]$ $[0, \pi]$



$\sin\theta = 0$

$\theta = 0, \pi, -\pi$

$\sin\theta = \frac{1}{2}$

$\theta = \frac{\pi}{6}, \frac{5\pi}{6}$

$\sin\theta = -1$

$\theta = -\frac{\pi}{2}$

Rejected

Question [JEE Mains-2022]

"Challenger"

Let $S = \left\{\theta \in [-\pi, \pi] - \left\{\pm\frac{\pi}{2}\right\} : \sin\theta\tan\theta + \tan\theta = \sin 2\theta\right\}$.

If $T = \sum_{\theta\in S} \cos 2\theta$, then $T + n(S)$ is equal to

$\hookrightarrow 4 + \hookrightarrow 5 = 9$ Ans.

$$S = \left\{0, \pi, -\pi, \frac{\pi}{6}, \frac{5\pi}{6}\right\}$$

$$\cos\frac{5\pi}{3} = \cos\left(2\pi - \frac{\pi}{3}\right) = \cos\frac{\pi}{3}$$

A. $7+\sqrt{3}$

B. 9

C. $8+\sqrt{3}$

D. 10

$$T = \sum_{\theta\in S} \cos 2\theta = \cos 0 + \cos 2\pi + \cos(-2\pi) + \cos\frac{2\pi}{3} + \cos 2\left(\frac{5\pi}{3}\right)$$

$$T = 1 + 1 + 1 + \frac{1}{2} + \frac{1}{2}$$

$$T = 4.$$

If $2\sin^3 x + \sin 2x \cos x + 4\sin x - 4 = 0$ has exactly 3 solutions in the interval $\left[0, \frac{n\pi}{2}\right], n \in N$, then the roots of the equation $x^2 + nx + (n-3) = 0$ belong to

A $(0, \infty)$

B $(-\infty, 0)$

C $\left(-\frac{\sqrt{17}}{2}, \frac{\sqrt{17}}{2}\right)$

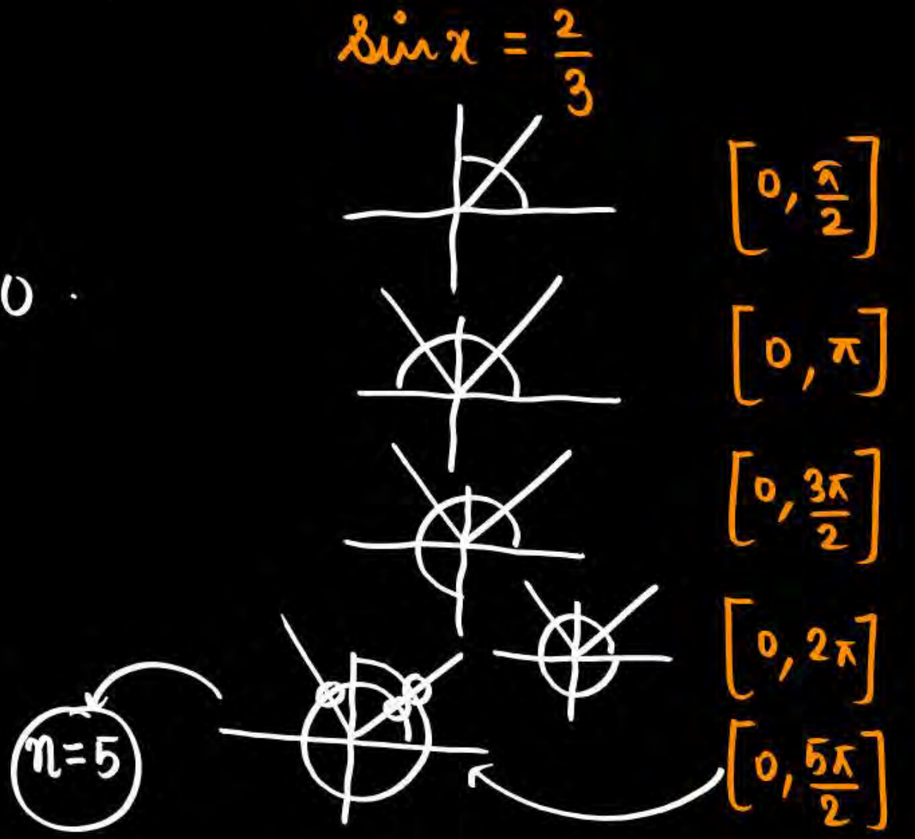
D Z

$2\sin^3 x + 2\sin x(1-\sin^2 x) + 4\sin x - 4 = 0$

$2\sin^3 x + 2\sin x - 2\sin^3 x + 4\sin x - 4 = 0$

$6\sin x = 4$

$\# \sin x = \frac{2}{3}$



Question [JEE Mains-2024]

If $2\sin^3 x + \sin 2x \cos x + 4\sin x - 4 = 0$ has exactly 3 solutions in the interval $\left[0, \frac{n\pi}{2}\right], n \in N$, then the roots of the equation $x^2 + nx + (n-3) = 0$ belong to

A $(0, \infty)$

B $(-\infty, 0)$

C $\left(-\frac{\sqrt{17}}{2}, \frac{\sqrt{17}}{2}\right)$

D Z

$n = 5$

$x^2 + 5x + 2 = 0$

$x = \frac{-5 \pm \sqrt{25 - 4\times 2}}{2}$

$x = \frac{-5 \pm \sqrt{17}}{2}$

$\frac{-5-\sqrt{17}}{2}$ -ve

$\frac{-5+\sqrt{17}}{2}$ -ve

Question

If $\cos 2\theta + 3\cos\theta = 0$, then

A $\theta = 2n\pi \pm \alpha$ where $\alpha = \cos^{-1}\left(\frac{\sqrt{17}-3}{4}\right)$

B $\theta = 2n\pi \pm \alpha$ where $\alpha = \cos^{-1}\left(\frac{-\sqrt{17}-3}{4}\right)$

C $\theta = 2n\pi \pm \alpha$ where $\alpha = \cos^{-1}\left(\frac{\pm\sqrt{17}-3}{4}\right)$

D None of these

$2\cos^2\theta - 1 + 3\cos\theta = 0$.

$2\cos^2\theta + 3\cos\theta - 1 = 0$.

$\cos\theta = \frac{-3 \pm \sqrt{9-4(2)(-1)}}{2\times 2}$.

$\cos\theta = \frac{-3 \pm \sqrt{17}}{4}$

$\frac{-3-\sqrt{17}}{4}$ ×

$\frac{-3+\sqrt{17}}{4}$ ✓

$\cos\theta = \frac{\sqrt{3}}{2}$

$\cos\theta = \cos\frac{\pi}{6}$

$\theta = 2n\pi \pm \frac{\pi}{6}$

$\cos\theta = \frac{\sqrt{17}-3}{4}$

$\cos\theta = \cos\alpha$

$\theta = 2n\pi \pm \alpha$,

$\cos\alpha = \frac{\sqrt{17}-3}{4}$

$\alpha = \cos^{-1}\left(\frac{\sqrt{17}-3}{4}\right)$

where .

Question [JEE Mains-2022]

Notes

The number of solutions of the equation

$\cos\left(x+\frac{\pi}{3}\right)\cos\left(\frac{\pi}{3}-x\right)=\frac{1}{4}\cos^2 2x, x \in [-3\pi, 3\pi]$ is:



A 8

B 5

C 6

D 7

DIFFERENT TYPES OF TRIGO. EQNs

Type-II : Solving Equation by introducing Auxiliary Argument

Angle.

A Trigonometric Equation is of the form $a\cos\theta \pm b\sin\theta = c$

M-1

$(\times)$ & $(\div)$ by $\sqrt{a^2+b^2}$

M-2

$$\cos\theta = \frac{1-\tan^2\frac{\theta}{2}}{1+\tan^2\frac{\theta}{2}}$$

$$\sin\theta = \frac{2\tan\frac{\theta}{2}}{1+\tan^2\frac{\theta}{2}}$$

Question

Find general value of x: $\sqrt{3}\cos x + \sin x = 2$.

$\sqrt{(\sqrt{3})^2 + 1^2} = 2$.

(M-1)

$$2\left(\frac{\sqrt{3}}{2}\cos x + \frac{1}{2}\sin x\right) = 2$$

$\frac{\sqrt{3}}{2} = \cos\frac{\pi}{6}$, $\frac{1}{2} = \sin\frac{\pi}{6}$.

$$\# \quad \cos\left(x - \frac{\pi}{6}\right) = 1 = \cos 0$$

(θ, α)

$$\Downarrow$$

$$x - \frac{\pi}{6} = 2n\pi \rightarrow \boxed{x = 2n\pi + \frac{\pi}{6},\ n \in I}$$

Question

Find general value of x: $\sqrt{3}\cos x + \sin x = 2$.

(M-2)

$$\sqrt{3} \times \left(\frac{1-\tan^2\frac{x}{2}}{1+\tan^2\frac{x}{2}}\right) + \left(\frac{2\tan\frac{x}{2}}{1+\tan^2\frac{x}{2}}\right) = 2$$

$$\sqrt{3}\left(1-\tan^2\frac{x}{2}\right) + 2\tan\frac{x}{2} = 2\left(1+\tan^2\frac{x}{2}\right)$$

$\tan\frac{x}{2} = t$

$$\sqrt{3} - \sqrt{3}t^2 + 2t = 2 + 2t^2$$

$$(2+\sqrt{3})t^2 - 2t + (2-\sqrt{3}) = 0.$$

$$(2+\sqrt{3})t^2 - 2t + (2-\sqrt{3}) = 0$$

$$t = \frac{2 \pm \sqrt{4-4\times 1}}{2(2+\sqrt{3})} = \frac{2}{2(2+\sqrt{3})} = 2-\sqrt{3}.$$

$$\tan\underbrace{\frac{x}{2}}_{\theta} = 2-\sqrt{3} = \tan\underbrace{\frac{\pi}{12}}_{\alpha}$$ ($\alpha = 15^\circ$)

$$\frac{x}{2} = n\pi + \frac{\pi}{12}$$

$$x = 2n\pi + \frac{\pi}{6}$$

Question

The general solution of the trigonometric equation $\sin x + \cos x = 1$ is given by



A $x = 2n\pi; n = 0, \pm 1, \pm 2, \ldots$

B $x = 2n\pi + \pi/2; n = 0, \pm 1, \pm 2, \ldots$

C $x = n\pi + (-1)^n \frac{\pi}{4} - \frac{\pi}{4}; n = 0, \pm 1, \pm 2, \ldots$

D None of the above

Challenger Question

Solve : $3\sqrt{3}\sin^3 x + \cos^3 x + 3\sqrt{3}\sin x \cos x = 1$

$3\sqrt{3}\sin^3 x + \cos^3 x - 1 = -3\sqrt{3}\sin x \cos x$.

$$(\sqrt{3}\sin x)^3 + (\cos x)^3 + (-1)^3 = 3(\sqrt{3}\sin x)(\cos x)(-1)$$

$a = b = c$ OR $a + b + c = 0$.

$\sqrt{3}\sin x = \cos x = -1$

$\sin x = -\frac{1}{\sqrt{3}}$ $\quad \cos x = -1$

n.p.

$\sqrt{3}\sin x + \cos x - 1 = 0$.

$\frac{\sqrt{3}}{2}\sin x + \frac{1}{2}\cos x = \frac{1}{2}$.

$\cos\left(x - \frac{\pi}{3}\right) = \cos\frac{\pi}{3}$ ✓

DO IT BY YOURSELF (DIBY)

DIBY-01

Solve : $\sqrt{3}\sec 2\theta = 2$.

[Ans. : $\theta = n\pi \pm \frac{\pi}{12}, n \in I$]

DIBY-02

Find general solution of equation $\cot x = \sqrt{3}$

(A) $x = n\pi + \frac{\pi}{3}$

(B) $x = n\pi + \frac{\pi}{6}$

(C) $x = n\pi + \frac{2\pi}{3}$

(D) $x = \frac{9\pi}{6}\frac{5\pi}{6}$

[Ans. : (B)]

DIBY-03

The number of values of x in the interval $[0, 5\pi]$ satisfying the equation $3\sin^2 x - 7\sin x + 2 = 0$ is:

(A) 0 (B) 2 (C) 6 (D) 8

[Ans. : (C)]

DIBY-04

General solution of the equation $(2\sin\theta - \sqrt{3})^2 + (\tan\theta + \sqrt{3})^2 = 0$

[Ans. : $(6n+2)\frac{\pi}{3}, n \in I$]

DIBY-05

Find the principal solution of $\sin 2x = -\frac{1}{2}$

(A) $\frac{7\pi}{6}, \frac{11\pi}{6}$

(B) $\frac{7\pi}{12}, \frac{11\pi}{12}, \frac{15\pi}{12}, \frac{19\pi}{12}$

(C) $\frac{7\pi}{12}, \frac{11\pi}{12}, \frac{19\pi}{12}, \frac{23\pi}{12}$

(D) None of these

[Ans. : (C)]

DIBY-06

Find the number of solution $\tan 3x = 1$ in the interval $x \in \left[-\frac{2\pi}{3}, \frac{5\pi}{6}\right]$.

[Ans. : 05]



Homework

Re-attempt all Class Examples

DPP & DIBY

It is never about end result

It's always about Journey

Thank You

#futureIITians